

Dhwanil Ranpura

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SKILLS

Languages: Python, R, SQL, JavaScript, TypeScript
Analytics & ML: EDA, Hypothesis Testing, Feature Engineering, XGBoost, Random Forest, Neural Networks, Scikit-learn, SHAP, Anomaly Detection, Time-Series Forecasting (ARIMA), Class Imbalance Handling, RAG, LLMs
Libraries: pandas, NumPy, matplotlib, seaborn, Plotly, FastAPI
BI & Visualization: Power BI, Tableau, Streamlit
Databases & Platforms: MySQL, SQLite, ChromaDB, AWS, Git, GitHub, Power Automate, Vercel
Frontend: React, Next.js, Tailwind CSS

EXPERIENCE

Data Automation Intern Sep 2024 – Feb 2025
Language & Culture Worldwide (LCW) Remote

- Built **Power BI** dashboards tracking operational KPIs across 10+ client initiatives, delivering structured analytical deliverables to non-technical stakeholders
- Automated data pipelines with **Power Automate**, reducing manual processing by **30%+** across multiple business systems
- Cleaned, transformed, and validated datasets from multiple business systems; surfaced trends to inform client strategy

PROJECTS

Commercial Lease Market Forecasting | *Python, R, ARIMA, Regression, EDA* | **1st Place – ASA DataFest** Apr 2025

- Analyzed **194K+** records to forecast commercial lease trends; built ARIMA + regression models achieving **92% accuracy** with 18% improvement via feature engineering
- Conducted EDA and hypothesis testing to surface macroeconomic drivers; presented client-ready findings to a cross-functional panel

SBA Loan Default Prediction | *Python, SQL, XGBoost, Random Forest, Neural Networks* Dec 2024

- Evaluated 4 classifiers on **899K financial records**; achieved **93% accuracy, 0.91 ROC-AUC** via cost-sensitive modeling and hyperparameter tuning
- Conducted profit-cost sensitivity analysis projecting **\$6.8M business impact**; delivered written and oral findings to non-technical stakeholders

Home Credit Risk Scorer | *Python, XGBoost, SHAP, Streamlit, Scikit-learn* Apr 2026

- Built two-layer credit scoring system on **300K+ loan applications**; XGBoost classifier (ROC-AUC 0.778) with class imbalance handling and threshold tuning
- Engineered interaction features (EXT_SOURCE × debt ratios) after identifying model scoring 267% DTI borrowers as Low Risk; implemented business rules engine with hard policy caps
- Applied **SHAP TreeExplainer** for per-applicant waterfall charts + global feature importance; backed by 16 unit tests

KPI Anomaly Detection System | *Python, Scikit-learn, Streamlit, FastAPI* Oct 2025

- Built end-to-end pipeline for real-time KPI anomaly detection with **FastAPI** backend for data ingestion, transformation, and validation
- Developed **Streamlit** dashboard with automated charting and trend reporting for non-technical users

SafeRoute – AI Safety Navigation | *React, Firebase, Python, Mapbox GL* | **3rd of 91 Teams – SJHacks** Apr 2025

- Trained supervised learning model on spatio-temporal crime data achieving **87% cross-validated accuracy**; built full-stack web app with real-time WebSocket alerts

LEADERSHIP

President, Spartan Analytics Dec 2025 – Present
San Jose State University San Jose, CA

- Lead 7-person executive team; delivered **Python and SQL workshops** to 30+ students and launched annual data competition for 30+ participants

EDUCATION

San Jose State University Expected Dec 2027
B.S. Data Science San Jose, CA